

Unit 12 Compositions/Inverses & Solving All Equations Assessment Study Guide

Unit 12 Compositions & Inverses of Functions

- ★ I can compose two functions.
- ★ I can use tables, graphs, and equations to evaluate compositions of functions.
- ★ I can determine whether a function has an inverse.
- ★ I can find inverse functions (linear, quadratic, radical, basic cubic), algebraically.
- ★ I can discuss what it means to graphically have inverse functions.
- ★ I can decompose a function to its parts.
- ★ I can prove functions are inverses of each other algebraically.
- ★ I can work with compositions of functions and inverses.

Solving all equations

- ★ I can determine if an equation is linear, quadratic, polynomial, radical, rational, or power.
- ★ I can solve each type of equation and check for extraneous when necessary.

1. Let $f(x) = x - 4$ and $g(x) = x^2 - x + 12$

$f(x) + g(x)$

$f(x) - g(x)$

$f(x) \cdot g(x)$

$\frac{g(x)}{f(x)}$

$f(g(x))$

$g(f(x))$

2. Let $f(x) = \frac{2}{x}$ and $g(x) = 2x^2 + 16x + 14$

$2f(4) + g(0)$

$xf(x) - g(x)$

$g(f(\frac{1}{2}))$

$f(g(x))$

Determine whether the inverse of f is a function. If it is, find the inverse function. If it is not, explain why.

3. $f(x) = 3x + 1$

4. $f(x) = 9x^2, x \geq 0$

5. $f(x) = \sqrt{\frac{x+1}{9}} - 3$

6. $f(x) = \frac{2x}{3} - 4$

7. $f(x) = \frac{3x+8}{4x+5}$

8. $f(x) = 5x^4$

Prove whether the functions are inverses or not.

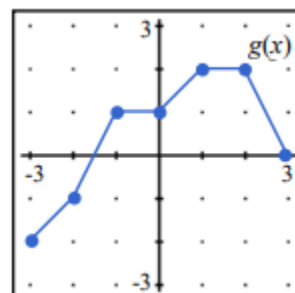
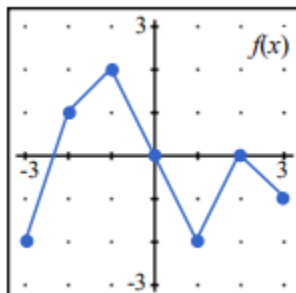
9. $f(x) = 2x - 9, g(x) = \frac{x}{2} + 9$

10. $f(x) = \frac{x-3}{4}, g(x) = 4x + 3$

11. $f(x) = \sqrt[5]{\frac{x+9}{5}}, g(x) = 5x^5 - 9$

12. $f(x) = 7x^{3/2} - 4, g(x) = (\frac{x+4}{7})^{3/2}$

Refer to the graphs of $f(x)$ and $g(x)$ to evaluate #s 13-18.



13. $f(g(1))$

14. $(f \circ g)(-3)$

15. $g(f(1))$

16. $(g \circ f)(-1)$

17. $f(f(3))$

18. $g(g(0))$

Refer to the tables of $H(t)$ and $r(t)$ to evaluate #s 19-24

t	1.0	1.5	2.0	2.5	3.0	3.5
$H(t)$	2.8	2.6	2.5	2.0	1.0	2.2

t	2.0	2.2	2.4	2.6	2.8	3.0
$r(t)$	1.2	1.5	3.0	2.8	2.5	2.0

19. $r(H(2.5))$

20. $(r \circ H)(1)$

21. $H(r(2.2))$

22. $(H \circ r)(3)$

23. $(H \circ H)(2)$

24. $(r(r(2.4)))$

Let $f(x) = x^2 + 2x - 3$, $g(x) = 3\sqrt{x}$, $h(x) = \sqrt{x^2 + 4}$, $j(x) = \frac{x-1}{x+1}$, $k(x) = (x + \sqrt[3]{x})^2$

25. $l(x) = 3\sqrt{x^2 + 2x - 3}$

26. $m(x) = (3\sqrt{x})^2 + 2(3\sqrt{x}) - 3$

27. $n(x) = \frac{\sqrt{x^2+4}-1}{\sqrt{x^2+4}+1}$

28. $p(x) = (x^2 + 2x - 3 + \sqrt[3]{x^2 + 2x - 3})^2$

29. $q(x) = 3\sqrt{x^2 + 4}$

30. $r(x) = \sqrt{(x^2 + 2x - 3)^2 + 4}$

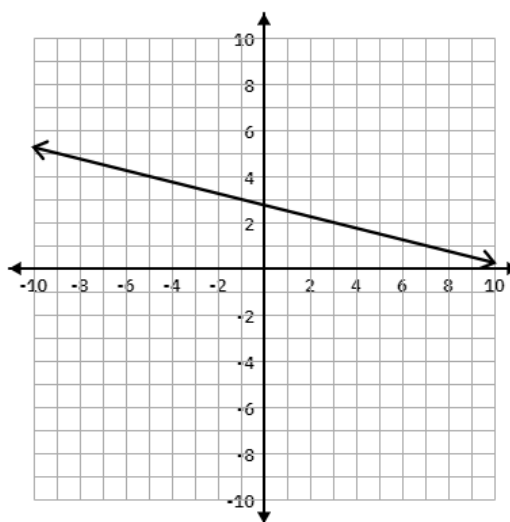
31. $s(x) = 3\sqrt{\frac{x-1}{x+1}}$

x	$f(x)$
2	4
3	2
4	3
7	7

$h(x) = 3x + 6$

$m(x) = \sqrt[3]{5x} + 1$

$Q(x)$



32. $f(f^{-1}(5))$

33. $Q^{-1}(3)$

34. $Q(3)$

35. $h^{-1}(m(25))$

36. $h(f^{-1}(4))$

KEY:

- 1) a) $(f + g)(x) = x^2 + 8$
b) $(f - g)(x) = -x^2 + 2x - 16$
c) $(f \cdot g)(x) = x^3 - 5x^2 + 16x - 48$
d) $(\frac{g}{f})(x) = x + 3 + \frac{24}{x-4}$
e) $f(g(x)) = x^2 - x + 8$
f) $g(f(x)) = x^2 - 9x + 32$
- 2) a) $2f(4) + g(0) = 15$
b) $xf(x) - g(x) = -2x^2 - 16x - 12$
c) $g(f(\frac{1}{2})) = 110$
d) $f(g(x)) = \frac{1}{(x+7)(x+1)}$
- 3) $f^{-1}(x) = \frac{1}{3}x - \frac{1}{3}$
- 4) $f^{-1}(x) = \frac{\sqrt{x}}{3}, x \geq 0$
- 5) $f^{-1}(x) = 9x^2 + 26$
- 6) $f^{-1}(x) = \frac{3}{2}x + 6$
- 7) $f^{-1}(x) = \frac{-5x+8}{4x-3}$
- 8) Does not have an inverse without a domain restriction
- 9) $f(g(x)) = g(f(x)) \rightarrow No$
- 10) $f(g(x)) = g(f(x)) \rightarrow Yes$
- 11) $f(g(x)) = g(f(x)) \rightarrow Yes$
- 12) $f(g(x)) = g(f(x)) \rightarrow Yes$
- 13) $f(g(1)) = -2$
- 14) $f(g(-3)) = 1$
- 15) $g(f(1)) = 1$
- 16) $g(f(-1)) = 2$
- 17) $f(f(3)) = 2$
- 18) $g(g(0)) = 2$
- 19) $r(H(2.5)) = 1.2$
- 20) $r(H(1)) = 2.5$
- 21) $H(r(2.2)) = 2.6$
- 22) $H(r(3)) = 2.5$
- 23) $H(H(2)) = 2.0$
- 24) $r(r(2.4)) = 2.0$
- 25) $l(x) = g(f(x))$
- 26) $m(x) = f(g(x))$
- 27) $n(x) = j(h(x))$
- 28) $p(x) = k(f(x))$
- 29) $q(x) = g(h(x))$
- 30) $r(x) = h(f(x))$
- 31) $s(x) = g(j(x))$
- 32) $f(f^{-1}(2)) = 2$
- 33) $q^{-1}(3) = -5$
- 34) $q(3) = 2$
- 35) $h^{-1}(m(25)) = 0$
- 36) $h(f^{-1}(4)) = 12$